

Residential Building

Energy Analysis and Simulation At
Brunnshög

GROUP 3

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AEBN11 - Residential Building - Integrating Thermal and Moisture Aspects

Department of Building and Environmental Technology

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Site Study

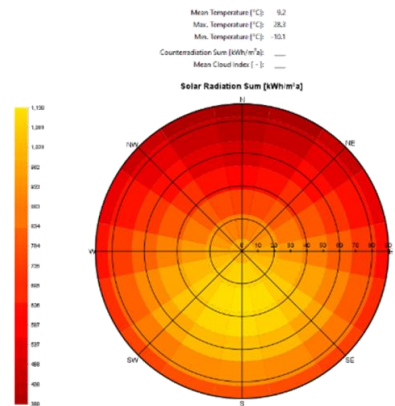
Introduction



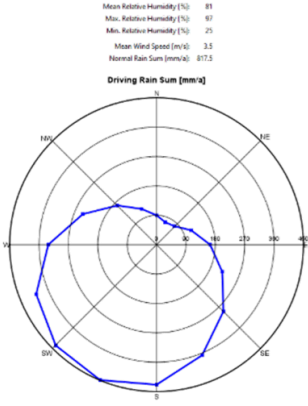
- Location: **Brunshög Lund Sweden**
- Project: **Multi-Store Residential Apartment**
- Neighbourhood: **Residences, Offices, Gardens..**
- Access: **Easy Access to Public by Transportation (Tram, Bus, Roads).**
- Site Condition: **Flat, Barren Terrain With Scattered Grass And Weeds.**
- Soil Type: **History Use As Fertile Agricultural Land (Rich Clay And Loam Content)**

Climate

Solar Radiation



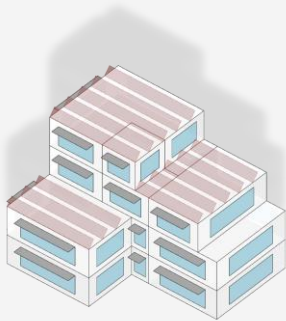
Wind Driven Rain (WDR)



- Average Annual Temperature: **9.2 °C**
- Hottest Annual Temperature: **28.3 °C** (June –Sept)
- Coldest Annual Temperature: **-10.1 °C** (December to February)
- Solar Latitude Angle: **46 ° South** (PV-Panels)
- Shading: **Onsite One Existing building in south (Design Hypothetical)**
- WDR: **South West 817.5 mm/a**

Initial Energy Study

- Energy Use Intensity (EUI) : 33.5 kWh/m^2 heated area per year
 - Heating Demand: 28.3 kWh/m^2 heated area per year
 - Proportion Beyond Comfort Threshold: 17% (under 26°C)
- Temperature range: 22.4°C to 29.2°C - From 1st Apr. to 30th Sep



Critic

Entrance: The door position, especially on the south side, should be redesigned as a room to reduce heating load and utilize natural heat efficiently for the indoor living room.

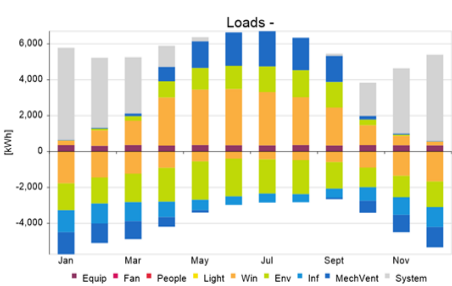
Orientation of Rooms: Room positions should be reorganized as they contribute to the overheating rate, exceeding the required limits.

PV Panel: PV panels on adjacent floors should be reconsidered as they occupy outdoor spaces like balconies and pose maintenance challenges.

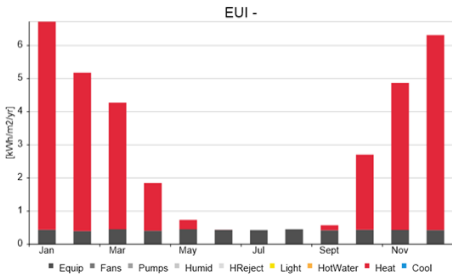
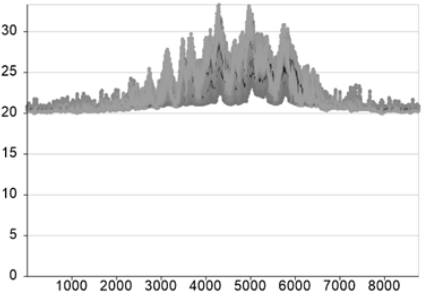
Understanding

Energy performance: EUI and heating load meeting the requirement.

Thermal comfort: The thermal comfort level interior do not reach the standard.

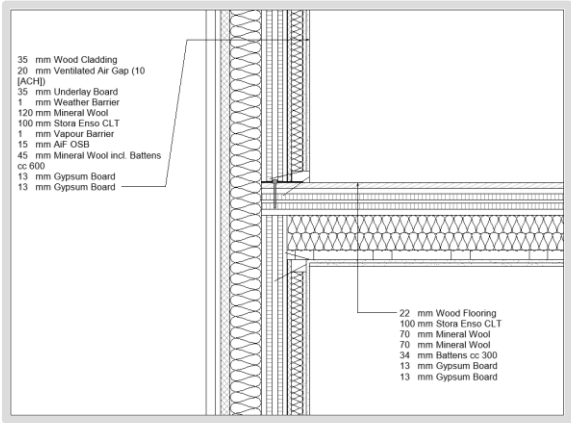


Zone Operative Temperature



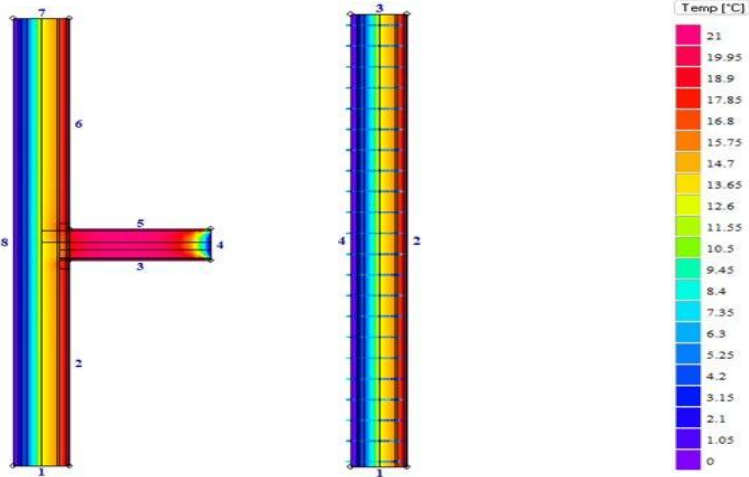
Initial Moisture Study

Wall with slab	Value
U-value	0.15 W/m ² K
Thickness	398 mm



Output	Value
RH < RH-critical	80%
Critical layer	Mineral wool layer
Mould index	Steady at 0
Mould growth rate	0 for all time periods
Thermal bridge	0.00186W/mK

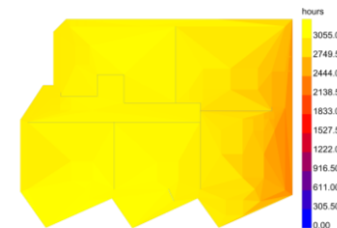
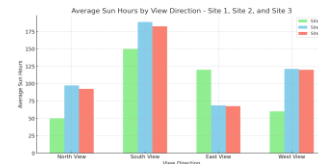
* Thermal bridge for wall with slab < 0.04 W/m·K



Site Selection And Building Form

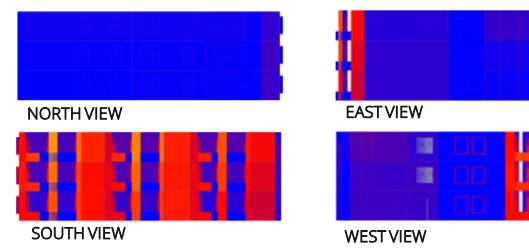


START



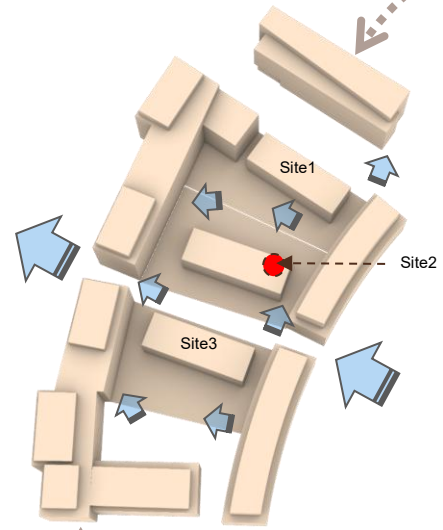
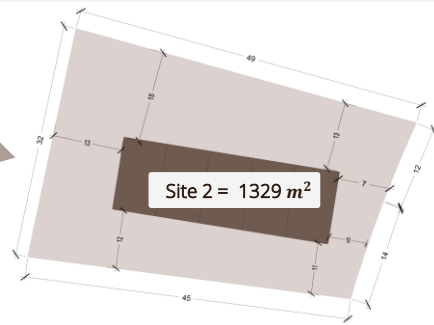
Site 2 is the optimal choice for maximizing sun exposure across all directions

Direct Sun Hour Analysis

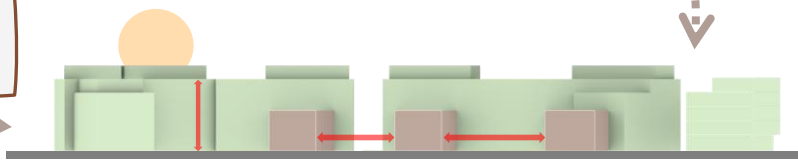


Building Form Facade Design Functional Zoning Roof Design Sustainability Features

Centralized Interaction Hub Visual Focal point Transition and Gathering Zones



Spatial Configuration Height Variation Solar Exposure Movement and Interaction Opportunities



Conclusion

- Focus on South-Facing Surfaces:**
 - Optimize for **solar gains** in winter.
 - Implement shading in summer to prevent overheating.
- Roof Design:**
 - Ideal for **solar panels** to take advantage of long summer days.
- East/West Facades:**
 - Use **shading solutions** to balance sunlight intensity.
- North Facade:**
 - Insulation is key to maintaining thermal comfort due to limited sunlight.

Floor Plans



TYPICAL FLOOR PLAN

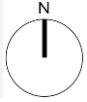


G-FLOOR AREA: 273.72m²
1st-FLOOR AREA: 273.72m²
2nd-FLOOR AREA: 273.72m²
TOTAL AREA: 821.18m²

WINDOW AREA: 211.28m²
FACADE AREA: 596.08m²

Scale : 1: 100

Elevation Views



North Elevation



East View



South View



West View



Elevation View



Isometric View



North West View



North West View



West View



East View



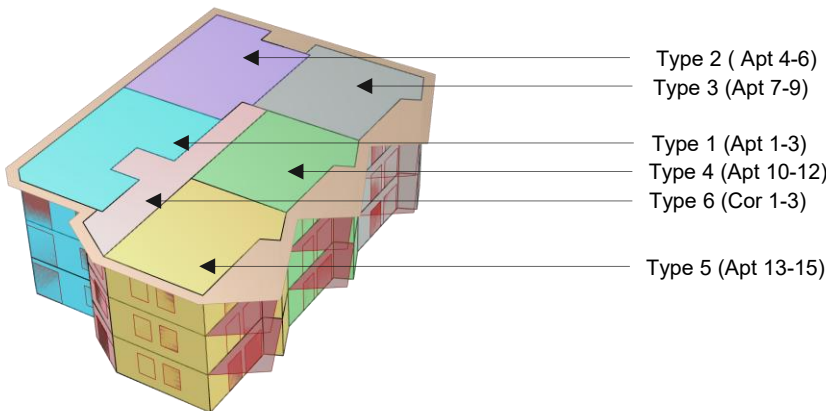
South View



North View

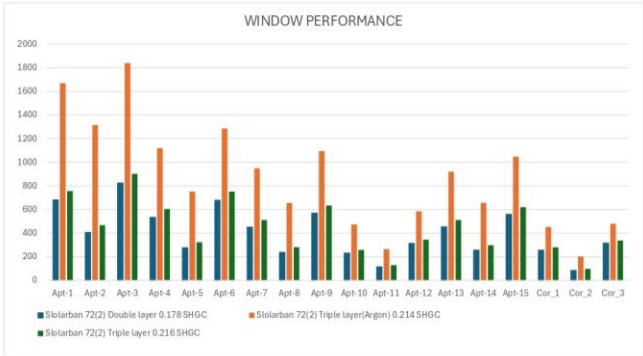
Energy Simulation Input

LOAD INPUT



(Types) Type of Apartment	1	2	3	4	5	Corridor
People Load (W/m²)	2.16	1.97	1.81	2.11	2.81	0
EL (W/m²)	2.4	2.4	2.4	2.4	2.4	2.4
Total	4.56	4.37	4.21	4.51	5.21	2.4

WINDOW



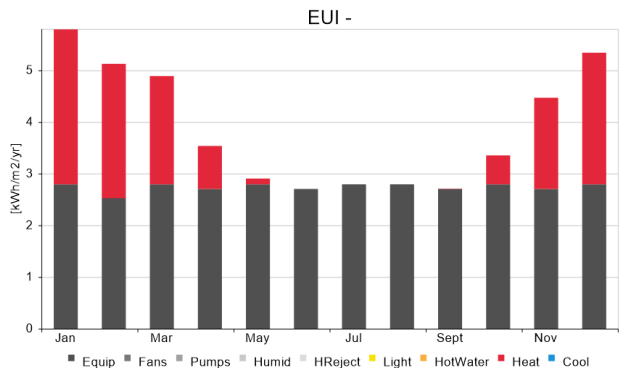
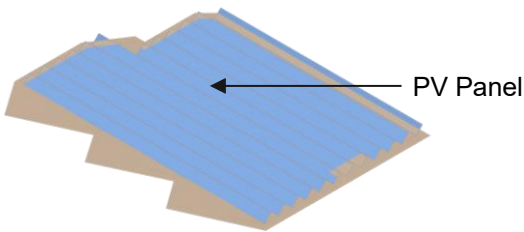
Glass Panel Type: Sololarban 72(2) Triple layer 0.216 SHGC

Building Envelope	U-value (W/m²K)
Roof	0.089
Wall	0.125
Ground	0.089
Slab	0.087

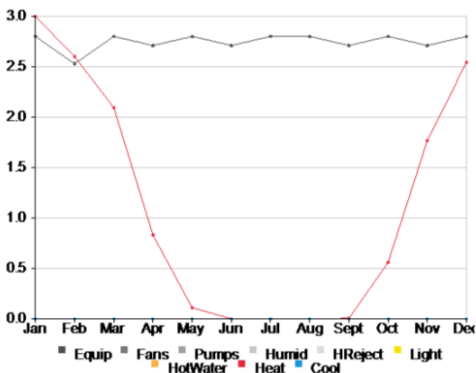
Parameter	Values
WWR	35%
U-value	0.57 W/m²K
SHGC	0.216
Tvis	0.512
Glazing Type	Triple
Gas	Air

Energy Performance & Thermal Comfort (Whole Building)

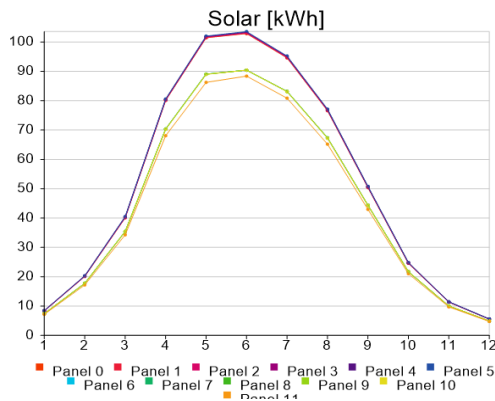
- HEATING DEMAND: **13.6 kWh/m² heated area per year**
- OVERHEATING PERCENTAGE (over 26°C): **2.4%**
- OPERATIVE TEMP: **20.43°C to 27.91°C**



ENERGY USE INTENSITY



HEATING DEMAND



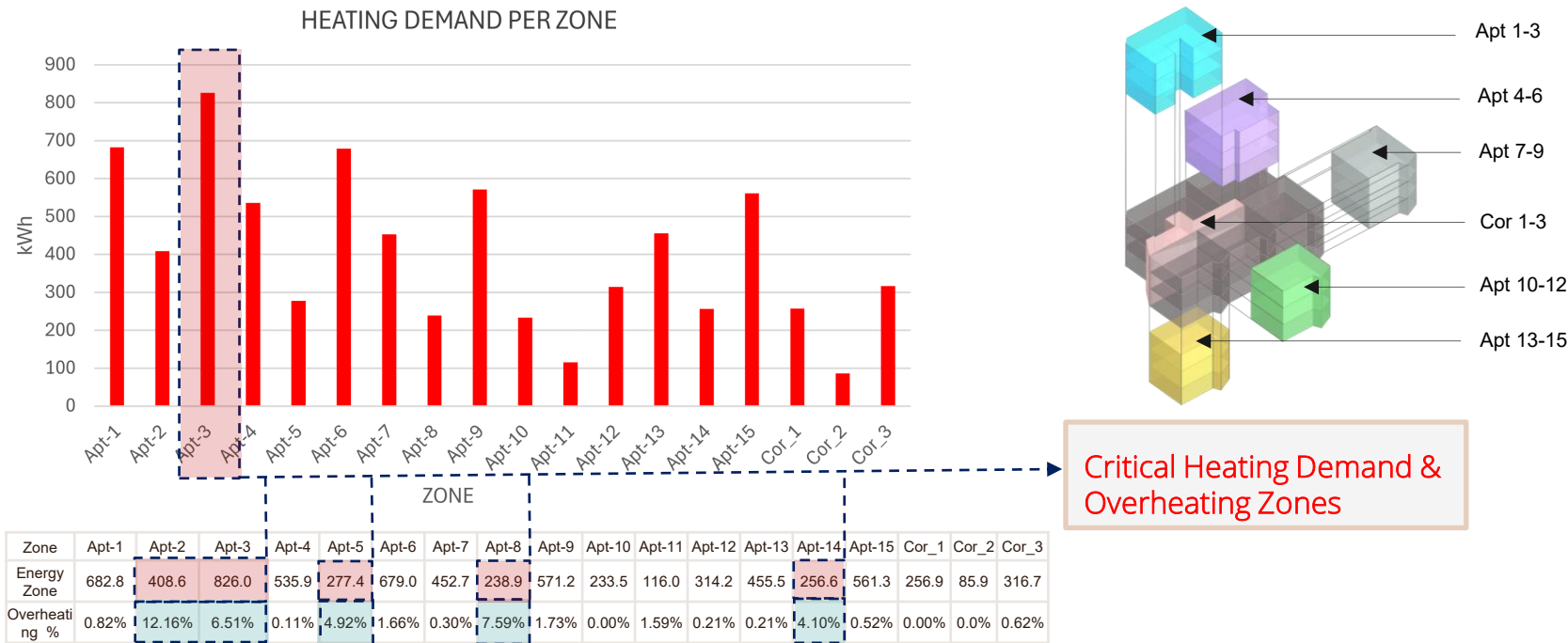
SOLAR ENERGY HARVESTING

COMPARE TO INTERMEDIATE DESIGN:

- HEATING LOAD: **Decreasing 52.3%**

- PV Panel Area : **294.09 m²**
- Panel Produced Energy: **29010 kWh**
- Total Energy Consumption: **32832 kWh Year**
- Energy Balance: **3822 kWh (88%)**

Energy Performance & Thermal Comfort (Zone Breakout)



- **Zone-Specific Heating Demand:** Apt-3 exhibits the highest energy consumption (above 700 kWh)
- **Low Demand Areas:** Apt-11 (Cor-1, Cor-2, Cor-3) shows the lowest heating demand, due to less enclosed spaces and lower heating requirements.
- **Overheating Zones:** Apt-2 exhibits an overheating percentage of 12.16%, which is above critical levels, pointing to potential inefficiencies in the heating system that require immediate attention

Energy Performance & Thermal Comfort

Strategy



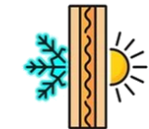
Optimize Shading: Use trees, awnings, or other shading devices to block direct sunlight, especially on windows and walls facing the sun. This reduces solar heat gain



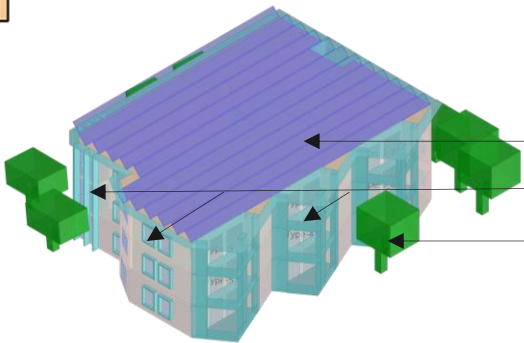
Optimize Shading: Implement solar panels to shade the roof surface, minimize heat absorption, and generate renewable energy, contributing to the goal of achieving net-zero emissions.



Optimize Shading: Implement solar panels to shade the roof surface and minimize heat absorption.



Insulated Wall: Implement the use of materials with high thermal mass.



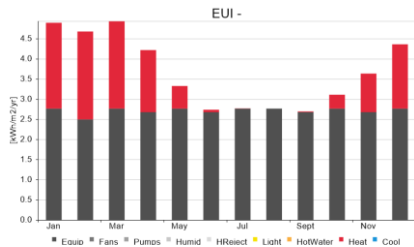
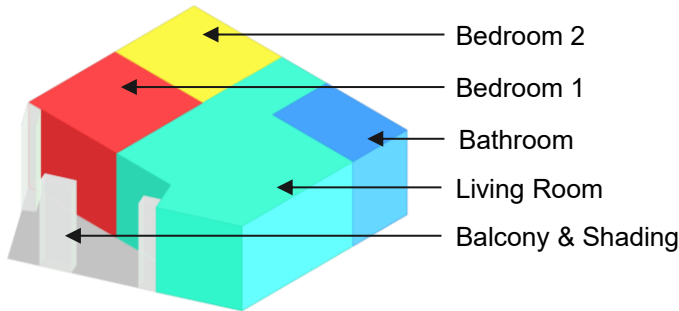
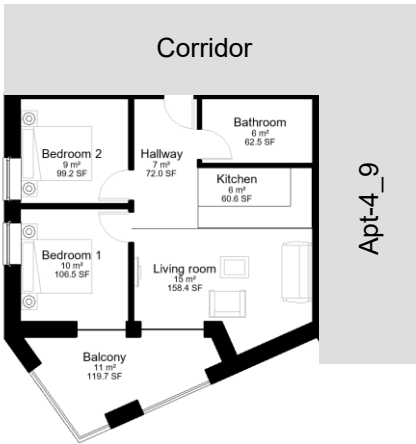
- PV Panel as Shading
- Facade as Shading
- Tree as Shading

Innovative Model

Zone Name	Building		Shading Device		Tree As Shading		Innovated Model	
	Energy Zone	Overheating %	Energy Zone	Overheating %	Energy Zone	Overheating %	Energy Zone	Overheating %
Apt-1	682.79	0.82	650.32	0.18	644.72	0.25	658.61	0.09
Apt-2	408.60	12.16	377.11	6.63	367.73	11.78	377.39	2.36
Apt-3	826.02	6.51	790.30	2.80	775.32	6.49	790.36	2.76
Apt-4	535.85	0.11	539.25	0.11	540.44	0.11	546.39	0.07
Apt-5	277.38	4.92	278.01	4.37	277.52	4.62	280.15	1.19
Apt-6	679.02	1.66	679.87	1.59	678.98	1.66	683.65	1.32
Apt-7	452.67	0.30	482.87	1.00	489.06	0.25	491.39	0.23
Apt-8	238.92	7.59	240.65	7.72	239.45	7.24	241.10	2.11
Apt-9	571.19	1.73	574.00	1.71	571.23	1.73	574.12	1.64
Apt-10	233.46	0.00	234.68	0.00	238.01	0.00	239.23	0.00
Apt-11	115.97	1.59	116.76	1.46	116.26	1.48	117.06	1.23
Apt-12	314.20	0.21	316.13	0.21	314.20	0.21	316.15	0.21
Apt-13	455.49	0.21	461.21	0.00	455.55	0.21	461.29	0.00
Apt-14	256.59	4.10	260.13	2.16	256.59	4.10	260.13	2.14
Apt-15	561.34	0.52	565.70	0.25	561.33	0.52	565.70	0.25
Cor_1	256.87	0.00	257.69	0.00	257.66	0.00	258.50	0.00
Cor_2	85.89	0.00	86.09	0.00	85.82	0.00	86.12	0.00
Cor_3	316.71	0.62	316.98	0.52	316.55	0.62	317.00	0.52

- Energy Use Intensity (EUI) : 46.7 kWh/m² heated area per year
- Heating load: 13.6 kWh/m² heated area per year

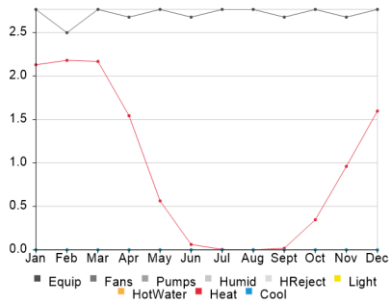
Energy Performance & Thermal comfort (GF Southwest Apartment)



ENERGY USE INTENSITY

- FLOOR AREA: **64m²**
- WINDOW AREA: **10m²**
- HEATING DEMAND: **11.57 kWh/m² heated area per year**
- HEATING ENERGY: **501.15kWh**
- OVERHEATING PERCENTAGE (over 26°C): **0%**
- ZONE OPERATIVE TEMP: **20.48°C to 23.79°C**

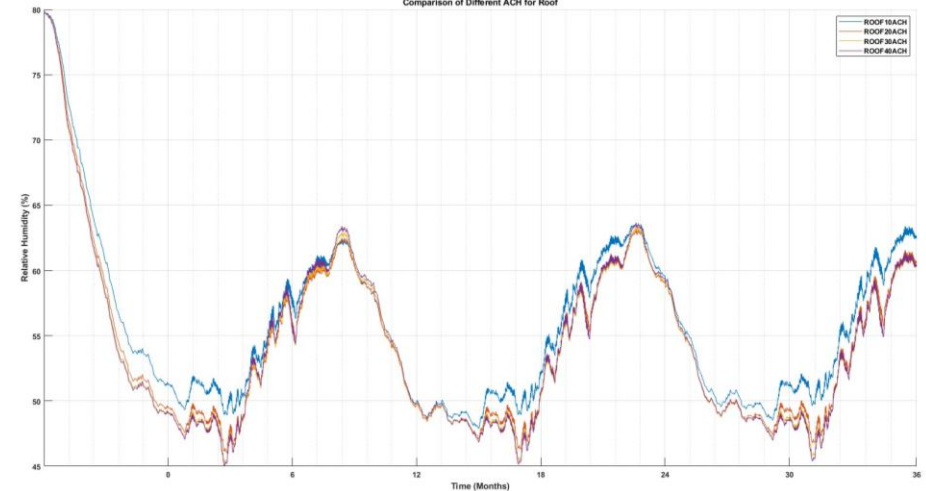
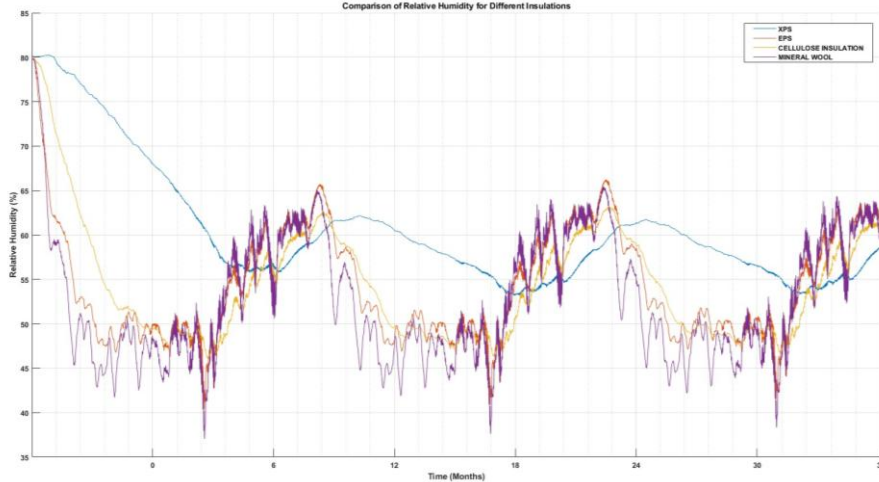
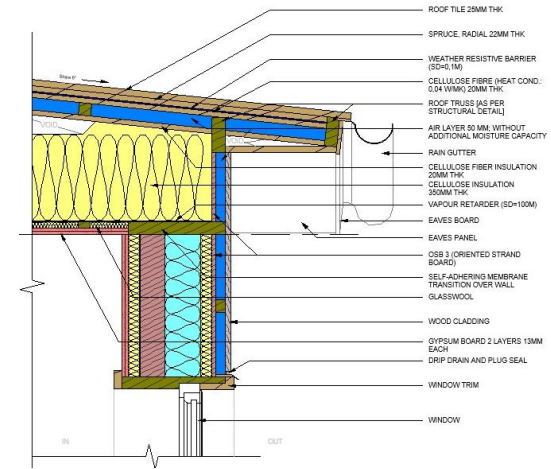
ZONES	AREA (m²)
Bedroom1	10
Bedroom2	9
Bath Room	6
Living Room	28
Balcony	11



HEATING LOAD

Moisture Safety Redesign - Roof

- 1% rain fraction on the exterior layer of the roof
 - Different insulations
(XPS, EPS, MINERAL WOOL & CELLULOSE INSULATION)
 - Cellulose Insulation & Different Air changes per hour
(10, 20, 30 & 40) [ACH]
- Combination of Cellulose insulation & 20 ACH – chosen



Moisture risk analysis (Roof) – Most critical layer of the Insulation

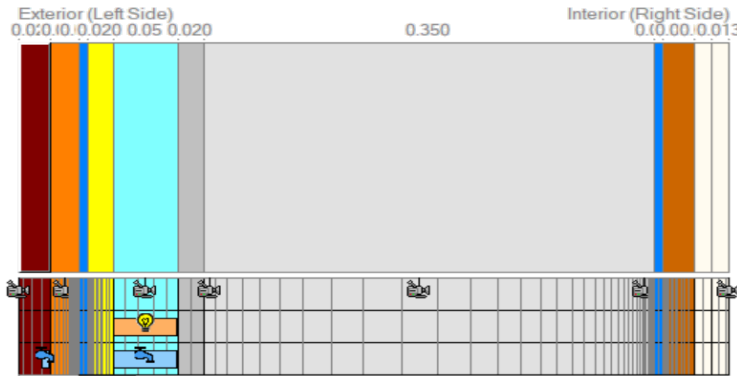


Figure 1 : WUFI setting

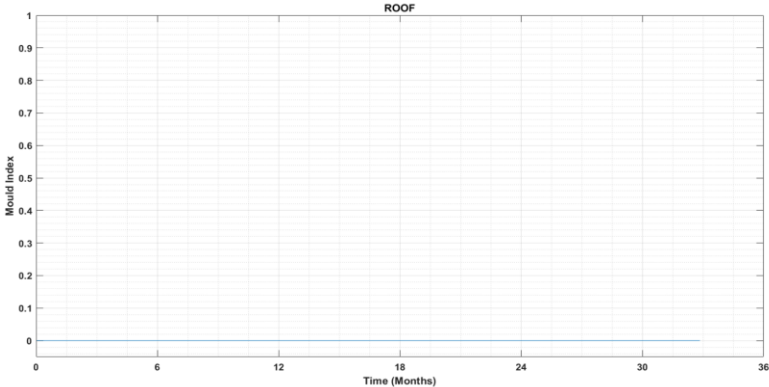


Figure 3: Mould index

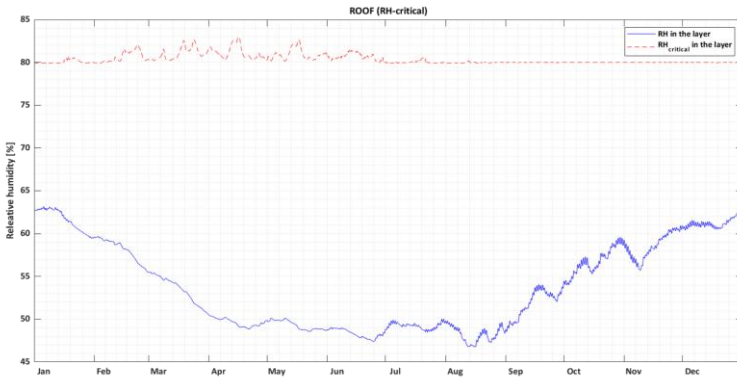


Figure 2: RH-critical

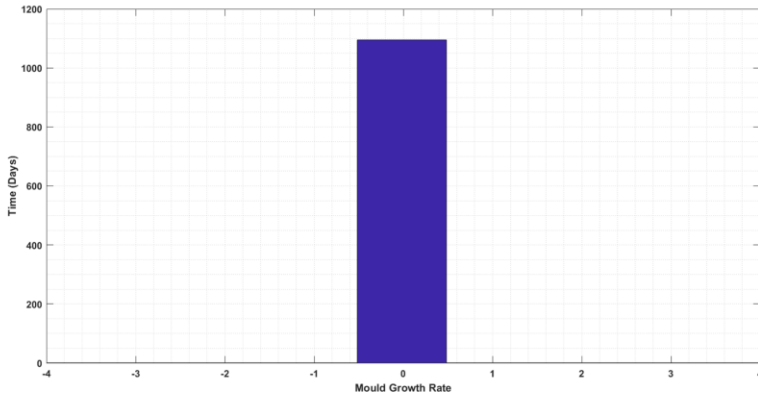
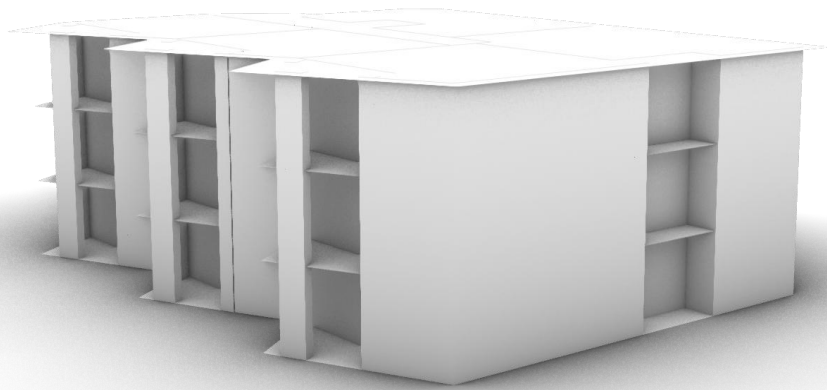


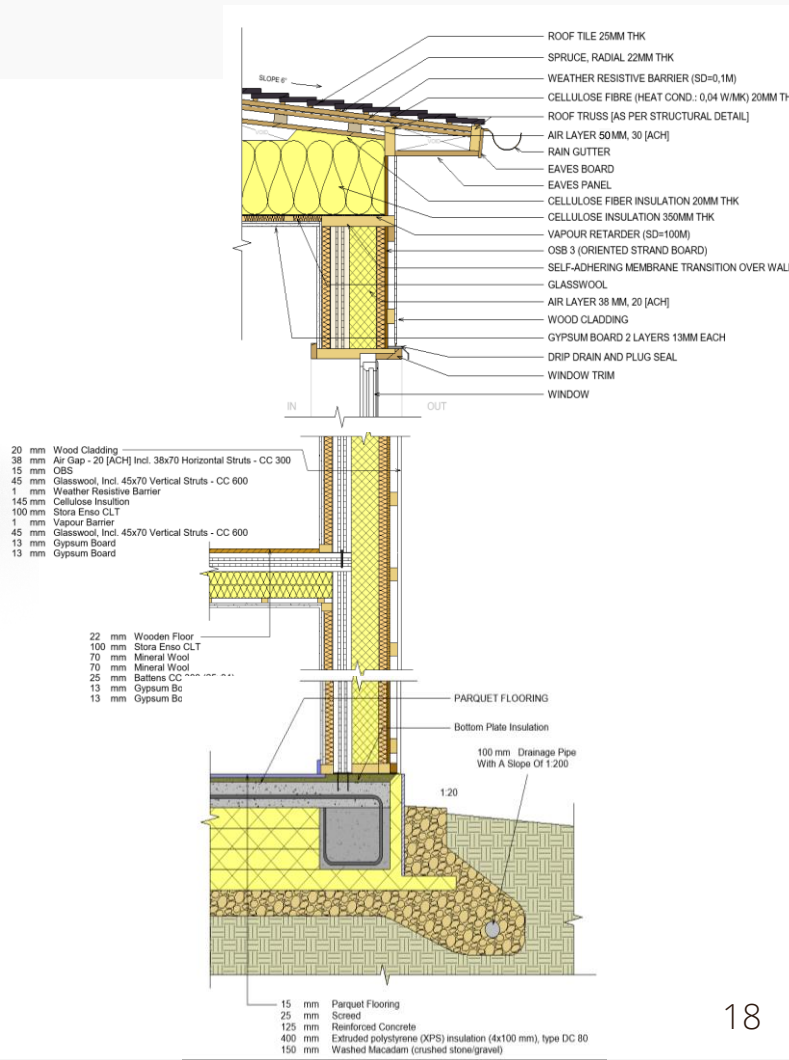
Figure 4 : Mould growth rate

Construction Details

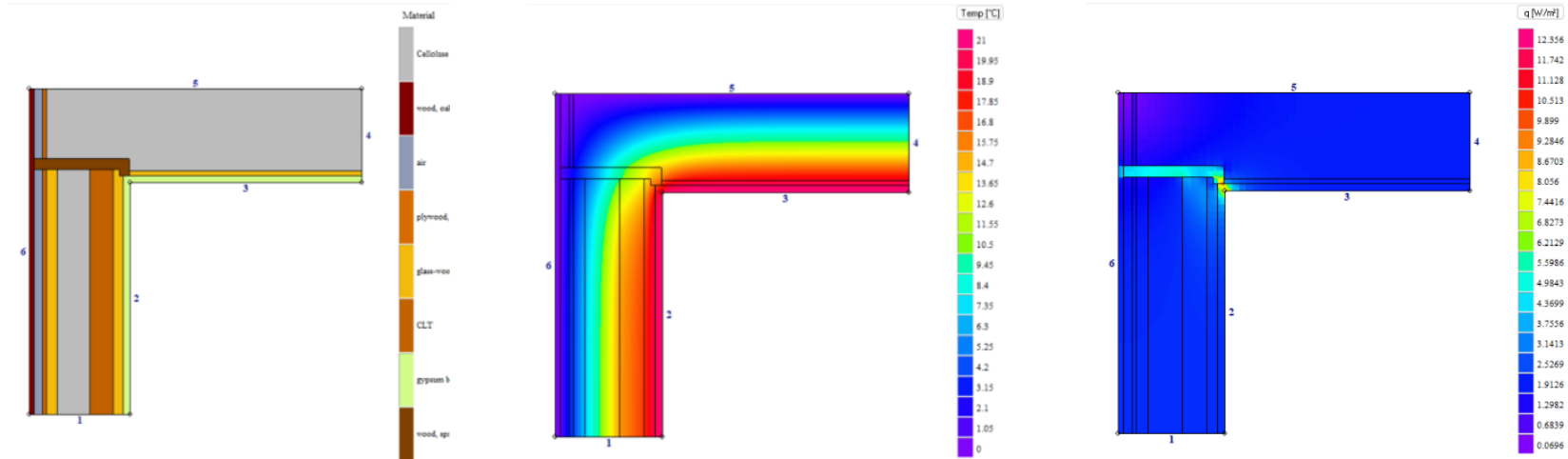


Compilation of different values of the building envelope

Building Envelope	Thickness (mm)	U-value (W/m ² K)	Thermal Bridge (W/mK)
Roof	540	0.089	0.03
Wall	438	0.125	0.0013
Ground	715	0.089	0.25
Slab	313	0.087	-

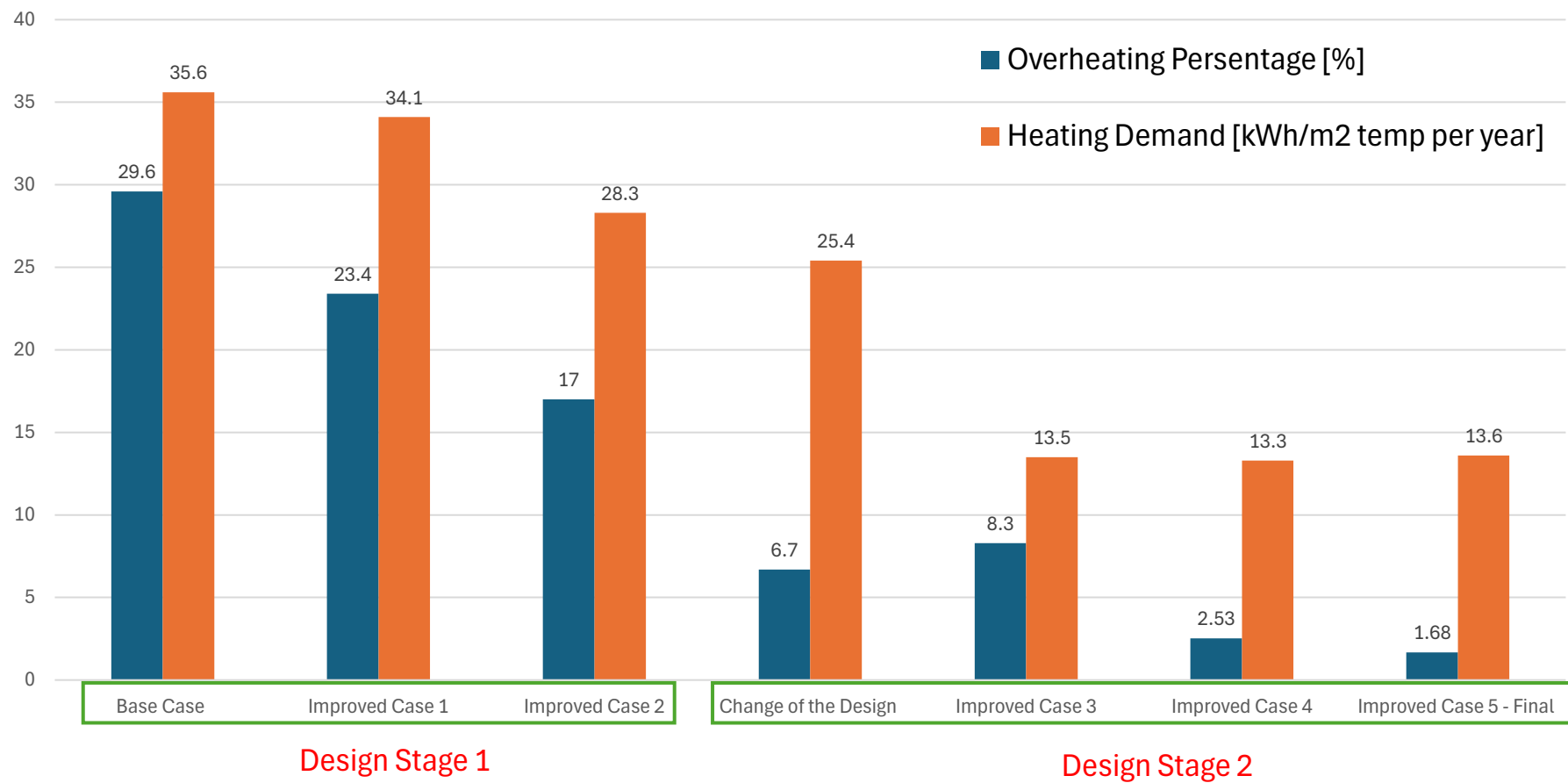


Thermal bridge (Roof)



Roof	Perfect wall	Perfect roof	Joint
q [W/m]	2.107	1.9158	4.7049
Temperature	$T_{out}=0^{\circ}\text{C}$ & $T_{in}=21^{\circ}\text{C}$		
Equation	$\Psi = (q_{joint} - (q_{roof} + q_{wall}))/\Delta T$		
Thermal bridge [W/mK]	0.03 < 0,8		

Project Progress





Thank You See you soon!!!